



## BOARD OF INTERMEDIATE & SECONDARY EDUCATION, LARKANO

*Established to serve affiliated schools and colleges across the Larkano region*

Higher Secondary School Certificate  
Examination Syllabus

# PHYSICS

Grades XI – XII

This syllabus will be examined in both Annual and Re-sit Examination sessions beginning with the Annual Examinations 2023.

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## Preface

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The Board of Intermediate & Secondary Education, Larkano (BISE Larkano) is a public examination body responsible for the conduct of Secondary School Certificate (SSC) and Higher Secondary School Certificate (HSSC) examinations for its affiliated schools and colleges. The Board is committed to developing examination syllabi that promote conceptual understanding and quantitative reasoning, and that remain aligned with the national curriculum and with contemporary international standards in science education.

This document sets out the syllabus for Physics at the HSSC level (Grades XI and XII). It has been developed in recognition of physics's role as a foundation for further study and employment in engineering, medicine, computing, the physical sciences and a wide range of other technical fields.

The syllabus was developed following consultation with subject specialists, teachers of affiliated institutions, and reviewers with expertise in physics education, and reflects a deliberate effort to move examination preparation away from rote memorisation of formulae and toward careful measurement, calculation and physical reasoning.

We are confident that this syllabus will support teachers and students of affiliated institutions in building a rigorous and practical understanding of physics, and we wish every success to those who study and teach it.

### **Controller of Examinations**

Board of Intermediate & Secondary Education, Larkano

## Understanding of BISE Larkano Syllabi

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1. The BISE Larkano syllabi guide students, teachers, parents and other stakeholders regarding the topics that will be taught and examined in each grade (XI and XII). In each syllabus document, the content progresses from simple to complex, facilitating gradual, conceptual learning.
2. The topics of each syllabus are divided into sub-topics and Student Learning Outcomes (SLOs). The sub-topics and SLOs define the depth and breadth at which each topic will be taught, learnt and examined.
3. Each SLO begins with an achievable, assessable command word such as ‘describe’, ‘apply’, ‘analyse’ or ‘evaluate’. Examination questions are framed using the same command words, or their equivalents, to elicit evidence of these competencies in students’ responses.
4. The SLOs are classified under three cognitive levels: Knowledge (K), Understanding (U), and Application and other higher-order skills (A), to support effective planning by teachers and the balanced construction of examination papers.
5. By focusing on the achievement of SLOs rather than on the reproduction of memorised facts, this syllabus aims to encourage students to reason scientifically, to justify their conclusions with evidence and calculation, and to communicate their thinking clearly — skills that extend well beyond a single assignment or examination.
6. While the syllabus recommends locally available textbooks and resources for achieving these outcomes, teachers and students are encouraged to draw on multiple teaching and learning resources, and to acknowledge any resource that materially shapes a solution submitted for assessment.

# Subject Rationale of Physics

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## What will you learn in Physics?

Physics is the study of matter, energy, and the fundamental forces and laws that govern their behaviour, from the motion of everyday objects to the structure of the atom. This syllabus teaches students to explain physical phenomena in terms of underlying quantities and laws, and to support their explanations with careful measurement and calculation rather than with memorised formulae.

Many students who can recall a formula have not yet learned to apply it — to work through a multi-step mechanics problem, to interpret a graph of experimental data, or to reason about what happens to a circuit when a component changes. This syllabus addresses that gap directly. Students will study mechanics, electricity and magnetism, waves and optics, and the foundations of modern physics, building a coherent picture of how the physical world behaves.

Throughout, the syllabus emphasises the application of physical principles to unfamiliar situations, and the ability to reason quantitatively from evidence — in the laboratory and in the examination room — rather than the recall of isolated formulae.

## Where will it take you?

The foundation provided by this syllabus prepares students for further study in Physics, Engineering, Computing and related disciplines, and for a wide range of careers, including:

- Mechanical / Electrical / Civil Engineer
- Physicist / Research Scientist
- Astronomer or Astrophysicist
- Medical Physicist / Radiologic Technologist
- Data Scientist
- Aerospace Engineer
- Meteorologist
- Electronics or Telecommunications Engineer

## How to approach the syllabus?

The concept map on the following page gives an overview of the entire syllabus. The topics and Student Learning Outcomes guide what has to be achieved, and the Scheme of Assessment explains how each topic will be weighted and examined.

## Concept Map

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The concept map below organises the syllabus into four connected strands, all resting on a foundation of Measurement, Mathematical Reasoning and Experimental Method — the quantitative and investigative skills that make every other strand of physics possible.

### Mechanics

- Measurements and Uncertainty
- Vectors and Equilibrium
- Kinematics, Force and Momentum
- Rotational and Circular Motion
- Work, Energy and Power

### Electricity and Magnetism

- Electrostatics
- Current Electricity and Circuits
- Magnetic Effects of Current
- Electromagnetic Induction

### Waves and Optics

- Simple Harmonic Motion
- Wave Motion and Sound
- Reflection and Refraction
- Interference, Diffraction and Polarisation

### Modern Physics

- The Bohr Model of the Atom
- Nuclear Physics and Radioactivity
- The Photoelectric Effect
- Mass-Energy Equivalence

*Measurement, Mathematical Reasoning and Experimental Method underpin all four strands: every calculation, every graph and every laboratory reading in this syllabus is ultimately a tool for reasoning carefully from evidence to a defensible conclusion.*

# Student Learning Outcomes of Physics

## Part I (Grade XI)

### 1. Measurements

| Topics and Sub-topics                    | Student Learning Outcomes                                                         | K | U | A |
|------------------------------------------|-----------------------------------------------------------------------------------|---|---|---|
| <b>1.1 Physical Quantities and Units</b> | describe the SI base quantities and their units;                                  | ✓ |   |   |
|                                          | express a derived unit in terms of SI base units;                                 |   |   | ✓ |
|                                          | use dimensional analysis to check the homogeneity of a physical equation;         |   |   | ✓ |
| <b>1.2 Measurement and Uncertainty</b>   | distinguish between systematic and random errors in a measurement;                |   | ✓ |   |
|                                          | express the result of a measurement with an appropriate estimate of uncertainty;  |   |   | ✓ |
|                                          | determine the number of significant figures appropriate to a calculated quantity; |   |   | ✓ |

### 2. Vectors and Equilibrium of Forces

| Topics and Sub-topics                           | Student Learning Outcomes                                                               | K | U | A |
|-------------------------------------------------|-----------------------------------------------------------------------------------------|---|---|---|
| <b>2.1 Vector Representation and Operations</b> | distinguish between a scalar quantity and a vector quantity;                            |   | ✓ |   |
|                                                 | resolve a vector into two perpendicular components;                                     |   |   | ✓ |
|                                                 | add two or more vectors using the head-to-tail method and by resolving into components; |   |   | ✓ |
| <b>2.2 Equilibrium</b>                          | state the conditions for the equilibrium of a body acted upon by coplanar forces;       | ✓ |   |   |
|                                                 | apply the conditions of equilibrium to solve a problem involving forces and torques;    |   |   | ✓ |

### 3. Motion, Force and Momentum

| Topics and Sub-topics                 | Student Learning Outcomes                                                                 | K | U | A |
|---------------------------------------|-------------------------------------------------------------------------------------------|---|---|---|
| <b>3.1 Kinematics</b>                 | describe motion in a straight line in terms of displacement, velocity and acceleration;   | ✓ |   |   |
|                                       | interpret a displacement-time or velocity-time graph to describe the motion of a body;    |   |   | ✓ |
|                                       | solve a problem involving uniformly accelerated motion using the equations of motion;     |   |   | ✓ |
| <b>3.2 Newton's Laws and Momentum</b> | state Newton's three laws of motion;                                                      | ✓ |   |   |
|                                       | apply Newton's second law to solve a problem involving force, mass and acceleration;      |   |   | ✓ |
|                                       | apply the principle of conservation of linear momentum to a collision between two bodies; |   |   | ✓ |

### 4. Rotational and Circular Motion

| Topics and Sub-topics          | Student Learning Outcomes                                                                             | K | U | A |
|--------------------------------|-------------------------------------------------------------------------------------------------------|---|---|---|
| <b>4.1 Circular Motion</b>     | describe uniform circular motion in terms of angular velocity and centripetal acceleration;           | ✓ |   |   |
|                                | apply the formula for centripetal force to solve a problem involving circular motion;                 |   |   | ✓ |
| <b>4.2 Rotational Dynamics</b> | define torque and moment of inertia;                                                                  | ✓ |   |   |
|                                | apply the relationship between torque, moment of inertia and angular acceleration to solve a problem; |   |   | ✓ |
|                                | apply the principle of conservation of angular momentum to a simple rotating system;                  |   |   | ✓ |

## 5. Work, Energy and Power

| Topics and Sub-topics                       | Student Learning Outcomes                                                                        | K | U | A |
|---------------------------------------------|--------------------------------------------------------------------------------------------------|---|---|---|
| <b>5.1 Work and Energy</b>                  | calculate the work done by a constant force acting on a body;                                    |   |   | ✓ |
|                                             | distinguish between kinetic energy and potential energy;                                         |   | ✓ |   |
|                                             | apply the work-energy theorem to solve a problem;                                                |   |   | ✓ |
| <b>5.2 Conservation of Energy and Power</b> | apply the principle of conservation of mechanical energy to a system in the absence of friction; |   |   | ✓ |
|                                             | calculate the power developed by a force or a simple machine;                                    |   |   | ✓ |

## Part II (Grade XII)

### 6. Electrostatics and Current Electricity

| Topics and Sub-topics          | Student Learning Outcomes                                                                     | K | U | A |
|--------------------------------|-----------------------------------------------------------------------------------------------|---|---|---|
| <b>6.1 Electrostatics</b>      | apply Coulomb's law to calculate the force between two point charges;                         |   |   | ✓ |
|                                | describe the concepts of electric field and electric potential;                               | ✓ |   |   |
|                                | apply Gauss's law to calculate the electric field of a simple, symmetric charge distribution; |   |   | ✓ |
| <b>6.2 Current Electricity</b> | describe Ohm's law and its limitations;                                                       | ✓ |   |   |
|                                | apply Kirchhoff's laws to analyse a simple direct-current circuit;                            |   |   | ✓ |
|                                | calculate the equivalent resistance of resistors connected in series and in parallel;         |   |   | ✓ |

### 7. Electromagnetism and Electromagnetic Induction

| Topics and Sub-topics                  | Student Learning Outcomes                                                            | K | U | A |
|----------------------------------------|--------------------------------------------------------------------------------------|---|---|---|
| <b>7.1 Magnetic Effects of Current</b> | describe the magnetic field produced by a current-carrying conductor;                | ✓ |   |   |
|                                        | calculate the force on a current-carrying conductor placed in a magnetic field;      |   |   | ✓ |
| <b>7.2 Electromagnetic Induction</b>   | state Faraday's law and Lenz's law of electromagnetic induction;                     | ✓ |   |   |
|                                        | calculate the electromotive force induced in a coil due to a changing magnetic flux; |   |   | ✓ |
|                                        | describe the principle of operation of a transformer or a simple generator;          |   | ✓ |   |

### 8. Waves and Oscillations

| Topics and Sub-topics             | Student Learning Outcomes                                                                   | K | U | A |
|-----------------------------------|---------------------------------------------------------------------------------------------|---|---|---|
| <b>8.1 Simple Harmonic Motion</b> | describe the characteristics of simple harmonic motion;                                     | ✓ |   |   |
|                                   | apply the equations of simple harmonic motion to a simple pendulum or a mass-spring system; |   |   | ✓ |
| <b>8.2 Wave Motion and Sound</b>  | distinguish between a transverse wave and a longitudinal wave;                              |   | ✓ |   |
|                                   | describe the phenomena of interference, diffraction and the Doppler effect;                 |   | ✓ |   |
|                                   | solve a problem involving the speed, frequency and wavelength of a wave;                    |   |   | ✓ |

### 9. Optics

| Topics and Sub-topics                | Student Learning Outcomes                                                                              | K | U | A |
|--------------------------------------|--------------------------------------------------------------------------------------------------------|---|---|---|
| <b>9.1 Reflection and Refraction</b> | apply the laws of reflection and refraction to construct a ray diagram for a mirror or a lens;         |   |   | ✓ |
|                                      | calculate the position and nature of an image formed by a mirror or a lens using the relevant formula; |   |   | ✓ |
| <b>9.2 Wave Optics</b>               | describe the phenomenon of interference of light with reference to Young's double-slit experiment;     |   | ✓ |   |

| Topics and Sub-topics | Student Learning Outcomes                                        | K | U | A |
|-----------------------|------------------------------------------------------------------|---|---|---|
|                       | describe the phenomena of diffraction and polarisation of light; | ✓ |   |   |

## 10. Modern Physics

| Topics and Sub-topics                       | Student Learning Outcomes                                                                             | K | U | A |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------|---|---|---|
| <b>10.1 Atomic and Nuclear Physics</b>      | describe the Bohr model of the hydrogen atom;                                                         | ✓ |   |   |
|                                             | calculate the energy released in a nuclear fission or fusion reaction using the mass-energy relation; |   |   | ✓ |
|                                             | describe the process of radioactive decay and interpret a decay curve;                                |   | ✓ |   |
| <b>10.2 Introduction to Quantum Physics</b> | describe the photoelectric effect and its significance for the particle nature of light;              |   | ✓ |   |
|                                             | apply the photoelectric equation to solve a numerical problem;                                        |   |   | ✓ |

## Scheme of Assessment

Each grade of Physics is examined through a Theory paper and a Practical (laboratory) examination, whose marks combine to the 100 marks recorded for the subject on the BISE Larkano Marks Certificate. The Theory paper carries 85 marks and the Practical examination carries 15 marks, matching the maximum marks shown against ‘Physics (TH)’ and ‘Physics (PR)’ for each year on the certificate.

### Grade XI

**Table 1: Number of Student Learning Outcomes by Cognitive Level**

| Topic No. | Topic                             | No. of Sub-topics | K   | U   | A   | Total |
|-----------|-----------------------------------|-------------------|-----|-----|-----|-------|
| 1         | Measurements                      | 2                 | 1   | 1   | 4   | 6     |
| 2         | Vectors and Equilibrium of Forces | 2                 | 1   | 1   | 3   | 5     |
| 3         | Motion, Force and Momentum        | 2                 | 2   | 0   | 4   | 6     |
| 4         | Rotational and Circular Motion    | 2                 | 2   | 0   | 3   | 5     |
| 5         | Work, Energy and Power            | 2                 | 0   | 1   | 4   | 5     |
|           | Total                             | 10                | 6   | 3   | 18  | 27    |
|           | Percentage                        |                   | 22% | 11% | 67% | 100%  |

**Table 2: Theory Exam Specification — Grade XI — Physics (TH)-I — 85 Marks**

| Topic No. | Topic                             | MCQs (Marks) | CRQs (Marks) | Total Marks |
|-----------|-----------------------------------|--------------|--------------|-------------|
| 1         | Measurements                      | 5            | 9 (1 CRQ)    | 14          |
| 2         | Vectors and Equilibrium of Forces | 6            | 10 (1 CRQ)   | 16          |
| 3         | Motion, Force and Momentum        | 7            | 13 (1 CRQ)   | 20          |
| 4         | Rotational and Circular Motion    | 6            | 11 (1 CRQ)   | 17          |
| 5         | Work, Energy and Power            | 6            | 12 (1 CRQ)   | 18          |
|           | Total                             | 30           | 55           | 85          |

**Table 2A: Overall Marks Distribution — Grade XI**

| Component             | Certificate Entry | Marks |
|-----------------------|-------------------|-------|
| Theory Examination    | Physics (TH)-I    | 85    |
| Practical Examination | Physics (PR)-I    | 15    |
| Total                 |                   | 100   |

## Grade XII

**Table 3: Number of Student Learning Outcomes by Cognitive Level**

| Topic No. | Topic                                          | No. of Sub-topics | K   | U   | A   | Total |
|-----------|------------------------------------------------|-------------------|-----|-----|-----|-------|
| 6         | Electrostatics and Current Electricity         | 2                 | 2   | 0   | 4   | 6     |
| 7         | Electromagnetism and Electromagnetic Induction | 2                 | 2   | 1   | 2   | 5     |
| 8         | Waves and Oscillations                         | 2                 | 1   | 2   | 2   | 5     |
| 9         | Optics                                         | 2                 | 1   | 1   | 2   | 4     |
| 10        | Modern Physics                                 | 2                 | 1   | 2   | 2   | 5     |
|           | Total                                          | 10                | 7   | 6   | 12  | 25    |
|           | Percentage                                     |                   | 28% | 24% | 48% | 100%  |

**Table 4: Theory Exam Specification — Grade XII — Physics (TH)-II — 85 Marks**

| Topic No. | Topic                                          | MCQs (Marks) | CRQs (Marks) | Total Marks |
|-----------|------------------------------------------------|--------------|--------------|-------------|
| 6         | Electrostatics and Current Electricity         | 7            | 13 (1 CRQ)   | 20          |
| 7         | Electromagnetism and Electromagnetic Induction | 6            | 11 (1 CRQ)   | 17          |
| 8         | Waves and Oscillations                         | 6            | 10 (1 CRQ)   | 16          |
| 9         | Optics                                         | 5            | 10 (1 CRQ)   | 15          |
| 10        | Modern Physics                                 | 6            | 11 (1 CRQ)   | 17          |
|           | Total                                          | 30           | 55           | 85          |

**Table 4A: Overall Marks Distribution — Grade XII**

| Component             | Certificate Entry | Marks |
|-----------------------|-------------------|-------|
| Theory Examination    | Physics (TH)-II   | 85    |
| Practical Examination | Physics (PR)-II   | 15    |
| Total                 |                   | 100   |

- A Multiple Choice Question (MCQ) requires candidates to choose one best/correct answer from four options. Each MCQ carries ONE mark.
- A Constructed Response Question (CRQ) requires students to respond with a structured derivation, a worked numerical problem, a labelled diagram, or an interpretation of experimental data, as appropriate to the SLO being assessed.
- There will be two Theory examinations, one at the end of Grade XI (Physics (TH)-I) and one at the end of Grade XII (Physics (TH)-II). Each Theory paper carries 85 marks, is of 3 hours' duration, and consists of Paper I (MCQs) and Paper II (CRQs).
- There will be two Practical examinations, one at the end of Grade XI (Physics (PR)-I) and one at the end of Grade XII (Physics (PR)-II). Each Practical examination carries 15 marks and requires students to carry out a measurement-based experiment under supervised laboratory conditions, and to record, graph and interpret their own results.

- The Theory and Practical marks for a given year are reported separately on the BISE Larkano Marks Certificate and are combined only in the year's subject total ( $85 + 15 = 100$ ). A candidate must attempt both components; a Theory or Practical mark cannot substitute for the other.
- Tables 1 and 3 indicate the number and nature of SLOs in each topic and are provided as a guide to the construction of examination papers; they do not translate directly into marks. Emphasis has deliberately been placed on Application and other higher-order skills, to discourage rote memorisation of formulae.
- Where a CRQ requires a numerical answer, full credit requires a correctly set-out method, the correct number of significant figures, and appropriate units, not merely the final value.

## Academic Integrity and Laboratory Safety

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All work submitted for assessment in Physics — whether a written answer, a calculation, or a laboratory report — must represent the individual, honest effort of the student submitting it. Affiliated institutions are expected to uphold the BISE Larkano Academic Integrity Policy in full, and to make students aware of the following subject-specific guidance.

### Collaboration and Use of Sources

Students are encouraged to discuss physical concepts, problem-solving strategies and experimental techniques with their peers, and to consult the prescribed textbook and other reputable resources to build their understanding. However, copying a solution, a calculation, or a laboratory report from any source — including another student, a book, or an online resource — and submitting it as one's own, with or without superficial changes, is a serious violation of academic integrity.

### Integrity of Experimental Data

Where practical work forms part of a student's assessment, all recorded readings and measurements must be genuine. Fabricating, adjusting or selectively omitting experimental data to obtain an expected or more convenient result is a serious violation of academic integrity, regardless of how small the discrepancy may seem; a result that disagrees with the expected value is itself valid data, and should be reported and discussed rather than altered.

### Laboratory and Apparatus Safety

Affiliated institutions are responsible for ensuring that all practical work is conducted under appropriate supervision, with attention to the safe use of electrical apparatus, optical equipment, and any other laboratory apparatus, and with students made aware of relevant hazard information before an activity begins. A student's disregard for reasonable safety instructions during practical work may be treated as a conduct matter independent of, and in addition to, their academic assessment.

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